

Colouring Sparse Oriented Graphs

Seminar Notes

Speaker: Alexander Clow (SFU)

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Primary textbook: Reinhard Diestel, *Graph Theory*, 6th ed.

Definitions first: ordinary colouring $\rightarrow$ homomorphisms $\rightarrow$ oriented colouring $\rightarrow$ sparse graphs $\rightarrow$ surfaces.

Compiled from the seminar abstract, the supplied screenshots and transcript, Diestel's textbook, and the cited research papers.

Contents

How to read these notes	3
1 Seminar overview	3
2 Basic definitions, with Diestel as the baseline	3
2.1 Graphs, adjacency, and independent sets	3
2.2 Graph homomorphisms	4
2.3 Proper colouring and chromatic number	4
2.4 Directed graphs, orientations, and tournaments	5
2.5 Oriented homomorphisms and oriented colourings	5
2.6 Directed distance and 2-dipath colouring	6
2.7 The dichromatic number	7
2.8 Fractional chromatic number	7
2.9 Why Shannon capacity appeared	7
3 Sparse graph parameters	8
3.1 Degree, average degree, and maximum average degree	8
3.2 Degeneracy and Diestel's colouring number	9
3.3 Why bounded degeneracy does not bound oriented colourings	9
4 Why naive greedy colouring fails	10
4.1 Ordinary greedy colouring is extension-stable	10
4.2 The oriented extension trap	10
5 Universal targets and full targets	10
5.1 Universal targets	10
5.2 Full oriented multipartite graphs	11
5.3 Why the target is multipartite	11
5.4 Why probabilistic full targets exist	12
6 Lower-bound methods	12
6.1 Local lower bounds from 2-dipaths	12
6.2 Global counting lower bounds	12
6.3 Universal-target obstructions	13
7 Fixed degeneracy	13

7.1	The square-root scale	13
7.2	Proof architecture described in the seminar	13
8	Graphs on surfaces	14
8.1	Embeddings, faces, Euler characteristic, and Euler genus	14
8.2	A Diestel-style comparison with ordinary colouring	14
8.3	Discrete curvature and discharging	15
9	The main theorem for Euler genus	15
9.1	Lower bound	15
9.2	Upper bound	16
10	What the seminar contributes to broader mathematical training	16
10.1	Homomorphisms replace labels by structure-preserving maps	16
10.2	Local solvability does not imply global compatibility	17
10.3	Different notions of sparsity answer different questions	17
10.4	Fractional relaxation links combinatorics and convex analysis	17
10.5	The probabilistic method is not Monte Carlo	17
10.6	Discharging is a discrete Gauss–Bonnet principle	18
11	A focused reading path in Diestel	18
12	Compact glossary	19
13	Final synthesis	20

How to read these notes

The seminar is not merely about adding arrows to an ordinary colouring problem. Its central question is whether structural sparsity still forces a small homomorphic target when a map must preserve both adjacency and direction. The notes therefore begin with precise definitions, establish the ordinary-colouring baseline from Diestel, and only then reconstruct the upper- and lower-bound strategies discussed in the talk.

Status of the transcript. The supplied automatic transcript contains several recognition errors. The two most important corrections are as follows. The dense-graph lower bound attributed to Wood is of order $\Omega(\sqrt{n})$, not $\Omega(n)$. The malformed expressions in the abstract should be read as exponential growth roughly $2^{\Delta/2}$ and the sparse-degeneracy bound $n^{1/2+o(1)}$. Statements described by the speaker as unpublished are marked accordingly.

1 Seminar overview

The talk follows the chain

ordinary colouring $\longrightarrow$ graph homomorphisms $\longrightarrow$ oriented colouring $\longrightarrow$ failure of naive greediness $\longrightarrow$

The opening motivation is that a proper colouring is exactly a homomorphism to a complete graph. Fractional colouring and Shannon capacity are then mentioned to illustrate the broader value of parameters that are monotone under graph homomorphisms. The main subject is the oriented chromatic number. The speaker compares it with the dichromatic number, explains why a standard degeneracy ordering does not directly support greedy extension, and introduces targets with strong local extension properties. The final part concerns two sparse regimes:

- graphs of fixed degeneracy, where the natural scale is approximately $\sqrt{n}$;
- graphs of Euler genus g , where the maximum oriented chromatic number lies between $\Omega(g/\log g)$ and $O(g \log g)$.

2 Basic definitions, with Diestel as the baseline

2.1 Graphs, adjacency, and independent sets

Definition 2.1: Graph and independent set

A finite simple graph is a pair

$$G = (V, E), \quad E \subseteq \binom{V}{2}.$$

The elements of V are vertices and the elements of E are edges. Vertices u, v are adjacent

when $uv \in E$. A set $I \subseteq V$ is independent if no two vertices of I are adjacent.

This is the language introduced in Diestel, Section 1.1, printed pages 2–3. Independent sets are fundamental in colouring because every colour class of a proper colouring is independent.

2.2 Graph homomorphisms

Definition 2.2: Graph homomorphism

A graph homomorphism from G to H is a map

$$\varphi : V(G) \rightarrow V(H)$$

such that

$$uv \in E(G) \implies \varphi(u)\varphi(v) \in E(H).$$

It need not be injective or surjective.

If two vertices have the same image under a homomorphism into a simple graph, they cannot be adjacent: otherwise their edge would be sent to a loop, and the target has no loops. Thus every fibre $\varphi^{-1}(x)$ is an independent set.

2.3 Proper colouring and chromatic number

Definition 2.3: Proper vertex colouring

A proper k -colouring is a map

$$c : V(G) \rightarrow [k] = \{1, \dots, k\}$$

such that adjacent vertices receive different colours. The chromatic number is

$$\chi(G) := \min\{k : G \text{ has a proper } k\text{-colouring}\}.$$

Proposition 2.4: Colouring is a homomorphism

For every graph G and positive integer k ,

$$\chi(G) \leq k \iff G \longrightarrow K_k.$$

Proof. A proper k -colouring becomes a vertex map into K_k by viewing the colours as the vertices of K_k . The endpoints of every edge have different colours, and every pair of distinct vertices of K_k is adjacent, so the map is a homomorphism. Conversely, if $\varphi : G \rightarrow K_k$ is a homomorphism, adjacent vertices cannot have the same image because K_k has no loops. Hence φ is a proper k -colouring. $\square$

This is precisely Exercise 5(i) of Diestel's Chapter 5, printed page 148. The first two seminar screenshots illustrate this equivalence: colour classes are the fibres of target vertices.

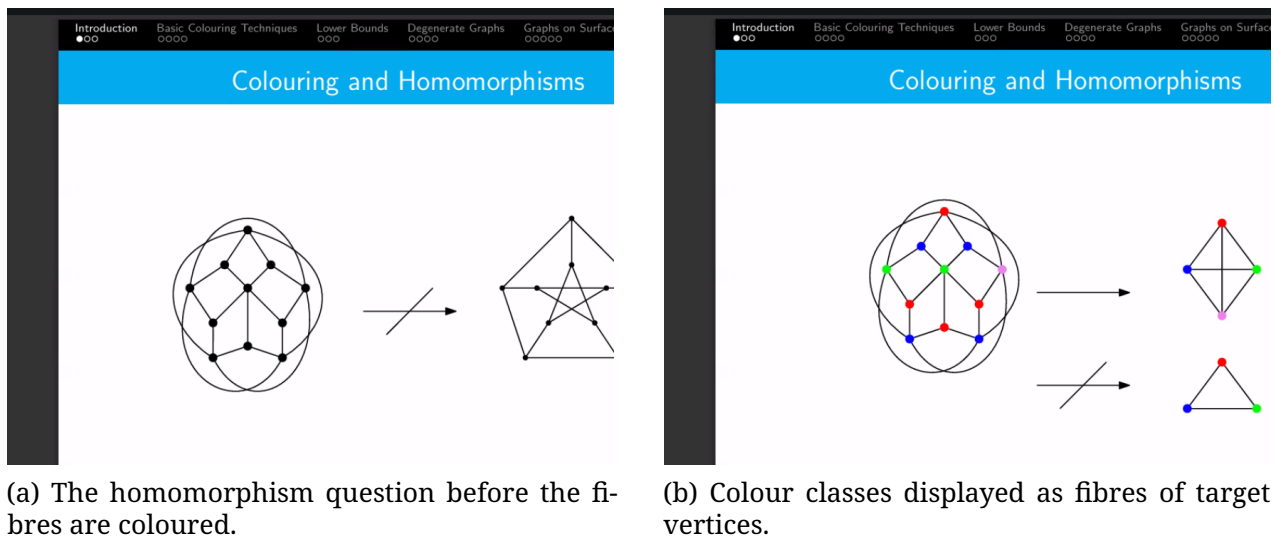


Figure 1: Seminar slides: ordinary colouring as a graph homomorphism.

2.4 Directed graphs, orientations, and tournaments

Definition 2.5: Digraph and oriented graph

A directed graph has arcs with an initial and a terminal vertex. An orientation of a simple graph is obtained by choosing exactly one of the two possible directions for each edge. An oriented graph is a digraph arising in this way; in particular, it has no loops and no pair of opposite arcs.

Diestel introduces these notions in Section 1.10, printed pages 27–28. The seminar concerns oriented graphs, rather than arbitrary digraphs that may contain both $u \rightarrow v$ and $v \rightarrow u$.

Definition 2.6: Tournament

A tournament is an orientation of a complete graph. Thus for every two distinct vertices x, y , exactly one of $x \rightarrow y$ and $y \rightarrow x$ is present.

Unlike the complete graph K_k , which is unique up to isomorphism, there are many non-isomorphic tournaments on k vertices. This is an early indication that oriented colouring is structurally richer than ordinary colouring.

2.5 Oriented homomorphisms and oriented colourings

Definition 2.7: Oriented homomorphism

An oriented homomorphism from an oriented graph D to an oriented graph H is a vertex map satisfying

$$u \rightarrow v \in A(D) \implies \varphi(u) \rightarrow \varphi(v) \in A(H).$$

It preserves adjacency and direction simultaneously.

Definition 2.8: Oriented chromatic number

For an oriented graph D ,

$$\chi_o(D) := \min\{|V(H)| : D \rightarrow H \text{ by an oriented homomorphism}\}.$$

For an undirected graph G ,

$$\chi_o(G) := \max\{\chi_o(D) : D \text{ is an orientation of } G\}.$$

Any oriented target can be completed to a tournament of the same order by orienting its missing edges arbitrarily. The original homomorphism remains valid. Hence one may equivalently minimise over tournament targets.

Equivalent colour-class formulation

An oriented colouring is a proper colouring with an additional consistency condition: between every pair of colour classes, all arcs must point in the same direction.

Equivalently, there cannot exist arcs $u \rightarrow v$ and $x \rightarrow y$ with

$$c(u) = c(y), \quad c(v) = c(x).$$

Otherwise the same two target colours would be forced to carry both opposite directions.

2.6 Directed distance and 2-dipath colouring

Definition 2.9: Directed distance and 2-dipath

The directed distance from u to v is the minimum length of a directed path from u to v , and is ∞ if no such path exists:

$$\text{dist}_D(u, v) := \min\{\ell : u = v_0 \rightarrow v_1 \rightarrow \cdots \rightarrow v_\ell = v\}.$$

A 2-dipath is a directed path of length two, $u \rightarrow x \rightarrow v$.

Definition 2.10: 2-dipath colouring

A 2-dipath colouring is a proper colouring in which the endpoints of every directed path of length two receive different colours. Its minimum number of colours is denoted by $\chi_2(D)$.

Lemma 2.11: Every oriented colouring is a 2-dipath colouring

Suppose $u \rightarrow x \rightarrow v$ and u, v receive the same colour a , while x receives colour b . The target would need both $a \rightarrow b$ and $b \rightarrow a$, impossible in an oriented graph. Therefore

$$\chi_2(D) \leq \chi_o(D).$$

This elementary local obstruction is one of the main ideas of the seminar. It gives lower

bounds and explains why ordinary greedy extension fails.

2.7 The dichromatic number

Definition 2.12: Dichromatic number

The dichromatic number of a digraph D is the least k for which

$$V(D) = V_1 \sqcup \cdots \sqcup V_k$$

and each induced subdigraph $D[V_i]$ is acyclic. We denote it by $\vec{\chi}(D)$.

The dichromatic number is a natural directed analogue of the ordinary chromatic number: a colour class is allowed to contain arcs, but not a directed cycle. It is not the parameter studied in the main theorem. For an orientation D of an undirected graph G ,

$$\vec{\chi}(D) \leq \chi(G) \leq \chi_o(D).$$

The first inequality holds because an independent set induces an acyclic digraph; the second holds because every oriented colouring is proper.

2.8 Fractional chromatic number

Definition 2.13: Fractional chromatic number

Let $\mathcal{I}(G)$ be the family of independent sets of G . Assign each independent set a nonnegative weight x_I . The fractional chromatic number is the optimum of

$$\chi_f(G) = \min \sum_{I \in \mathcal{I}(G)} x_I$$

subject to

$$\sum_{I \ni v} x_I \geq 1 \quad (v \in V(G)), \quad x_I \geq 0.$$

Ordinary colouring is an integral covering of the vertex set by independent colour classes. Fractional colouring relaxes the integral choice to nonnegative real weights, so it is a linear-programming relaxation. In particular,

$$\omega(G) \leq \chi_f(G) \leq \chi(G).$$

2.9 Why Shannon capacity appeared

The Shannon capacity $\Theta(G)$ measures the asymptotic size of zero-error codes and may be written in terms of independence numbers of strong graph powers:

$$\Theta(G) := \sup_{r \geq 1} \alpha(G^{\boxtimes r})^{1/r}.$$

The slide is titled "Colouring and Homomorphisms". It has a navigation bar at the top with five items: "Introduction" (selected), "Basic Colouring Techniques", "Lower Bounds", "Degenerate Graphs", and "Graphs on Surfaces". Below the title, it says "An example:". Then, under a "Definition" header, it defines the fractional chromatic number $\chi_f(G)$ as the minimum of $\sum_{I \in \mathcal{I}(G)} x_I$ subject to $\sum_{v \in I} x_I \geq 1$ for all $v \in V$ and $x_I \geq 0$ for all $I \in \mathcal{I}(G)$.

Introduction ●○○ Basic Colouring Techniques ○○○○ Lower Bounds ○○○ Degenerate Graphs ○○○○ Graphs on Surfaces ○○○○

Colouring and Homomorphisms

An example:

Definition

Let $G = (V, E)$ be a graph, and let $\mathcal{I}(G)$ denote the collection of all independent sets of G . For each $I \in \mathcal{I}(G)$, introduce $x_I \geq 0$. Then

$$\chi_f(G) = \min \sum_{I \in \mathcal{I}(G)} x_I$$

subject to

$$\sum_{v \in I} x_I \geq 1 \quad \text{for all } v \in V,$$

$$x_I \geq 0 \quad \text{for all } I \in \mathcal{I}(G)$$

Figure 2: The fractional-colouring linear programme shown in the seminar.

The speaker did not use it in the proof of the main theorem. It served as motivation for the homomorphism viewpoint: difficult graph parameters can sometimes be recovered as pointwise minima of better-behaved parameters that are monotone under graph homomorphisms. This is an example of a broader principle—understanding all structure-preserving monotone functionals can clarify a parameter that is otherwise difficult to compute directly.

3 Sparse graph parameters

3.1 Degree, average degree, and maximum average degree

Definition 3.1: Degree parameters

For a graph G ,

$$\delta(G) := \min_{v \in V(G)} d(v), \quad \Delta(G) := \max_{v \in V(G)} d(v).$$

The maximum average degree is

$$\text{mad}(G) := \max_{\emptyset \neq H \subseteq G} \frac{2|E(H)|}{|V(H)|}.$$

Maximum average degree inspects every subgraph, so a dense core cannot be hidden inside a large sparse remainder.

3.2 Degeneracy and Diestel's colouring number

Definition 3.2: Degeneracy

A graph is d -degenerate if every nonempty subgraph has a vertex of degree at most d . Its degeneracy is

$$d(G) := \max_{\emptyset \neq H \subseteq G} \delta(H).$$

A graph is d -degenerate if and only if its vertices can be ordered so that every vertex has at most d earlier neighbours. One obtains such an order by repeatedly deleting a current vertex of degree at most d and reversing the deletion order.

Diestel formalises the greedy parameter in Section 5.2. His colouring number satisfies

$$\chi(G) \leq \text{col}(G) = 1 + \max_{H \subseteq G} \delta(H) = d(G) + 1.$$

Thus bounded degeneracy gives an immediate constant upper bound for ordinary colouring. Moreover,

$$d(G) \leq \text{mad}(G) < 2d(G)$$

for nonempty graphs with $d(G) \geq 1$, so degeneracy and maximum average degree are equivalent up to a factor of two.

3.3 Why bounded degeneracy does not bound oriented colourings

Example 3.3: One-subdivision of a clique

Start with K_r and subdivide each edge once. The resulting graph $S(K_r)$ is 2-degenerate. Indeed, any subgraph containing a subdivision vertex has a vertex of degree at most two, while a subgraph consisting only of original vertices has no edges.

Orient the original complete graph as a tournament. If an original edge is oriented $u \rightarrow v$, orient its subdivided path as

$$u \longrightarrow x_{uv} \longrightarrow v.$$

Every pair of original vertices is now the pair of endpoints of a directed path of length two, so all original vertices require distinct colours in every oriented colouring. Since

$$N = |V(S(K_r))| = r + \binom{r}{2},$$

we obtain

$$\chi_o(S(K_r)) \geq r = \Theta(\sqrt{N}).$$

This is the central contrast:

$$2\text{-degenerate} \implies \chi(G) \leq 3,$$

but the oriented chromatic number is not bounded by any constant and can already grow on the square-root scale.

4 Why naive greedy colouring fails

4.1 Ordinary greedy colouring is extension-stable

In ordinary colouring, once a partial proper colouring has been constructed, a new vertex only forbids the colours used by its coloured neighbours. If at most d neighbours precede it in a degeneracy order, one of $d + 1$ colours remains available. Future vertices cannot retroactively invalidate an earlier legal choice.

4.2 The oriented extension trap

Suppose two already-coloured vertices u, w receive the same colour a . At the current stage they may be nonadjacent, so there is no visible conflict. A later vertex v may create

$$u \longrightarrow v \longrightarrow w.$$

If v receives colour b , the target must contain both $a \rightarrow b$ and $b \rightarrow a$. The partial colouring cannot be extended, regardless of how many unused colours are available for v .

Core obstruction

Ordinary colouring constraints are essentially one-step constraints. Oriented colouring constraints propagate along directed paths of length two. Consequently, a future vertex can expose an incompatibility between two earlier colour choices.

A successful argument must therefore maintain more information than the list of colours present in the current neighbourhood. It must control the directed distance-two relations that can later force opposite requirements between the same colour classes.

5 Universal targets and full targets

5.1 Universal targets

Definition 5.1: Universal target

An oriented graph H is a universal target for a class $\mathcal{C}$ if

$$D \longrightarrow H \quad \text{for every } D \in \mathcal{C}.$$

A standard greedy proof for an entire graph class often proves more than a numerical upper bound: it produces one fixed target to which every graph in the class maps.

Lemma 5.2: Disjoint-union principle

Suppose $\mathcal{C}$ is closed under disjoint unions and every graph in $\mathcal{C}$ has oriented chromatic number at most k . Then $\mathcal{C}$ has a universal target of order at most k .

Proof sketch. There are only finitely many isomorphism types of oriented graphs on

at most k vertices, say $H_1, \dots, H_t$. If none is universal, choose $D_i \in \mathcal{C}$ with $D_i \not\rightarrow H_i$. Their disjoint union belongs to $\mathcal{C}$, hence maps to some H_j . Restricting the map to D_j gives the contradiction $D_j \rightarrow H_j$. $\square$

This principle also creates lower bounds: different examples may force incompatible structural properties in any universal target, and their disjoint union forces all those properties simultaneously.

5.2 Full oriented multipartite graphs

Definition 5.3: Orientation pattern

For an ordered tuple $U = (u_1, \dots, u_t)$ and a vertex x , record the direction between x and each u_j by

$$F(U, x, H)_j = \begin{cases} 1, & x \rightarrow u_j, \\ -1, & u_j \rightarrow x. \end{cases}$$

Definition 5.4: (k, d, N) -full graph

Let H be a complete oriented k -partite graph with parts $P_1, \dots, P_k$, each of order N . It is (k, d, N) -full if, for every part P_i , every ordered tuple U of at most d vertices outside P_i , and every sign pattern $\varepsilon \in \{-1, 1\}^{|U|}$, there exists $x \in P_i$ satisfying

$$F(U, x, H) = \varepsilon.$$

Thus every prescribed small in/out pattern can be realised by a vertex in any designated part.

5.3 Why the target is multipartite

The seminar describes a product-colouring strategy.

1. First assign a coarse 2-dipath colour to every source vertex.
2. Associate each coarse colour i with a part P_i of the target.
3. The final oriented colour of a vertex is a specific target vertex chosen inside the prescribed part.
4. The full property supplies a target vertex with the correct directions toward all previously mapped neighbours.

The coarse colouring prevents dangerous directed distance-two pairs from being placed in the same part. The fine choice within a part then realises the required local orientation pattern. This is the bookkeeping device missing from naive greedy colouring.

5.4 Why probabilistic full targets exist

Orient independently and uniformly every edge between distinct parts. Fix a tuple of length $t \leq d$ and a prescribed sign pattern. A particular candidate vertex realises that pattern with probability 2^{-t} . The probability that all N candidates in one part fail is at most

$$(1 - 2^{-t})^N \leq e^{-N/2^t}.$$

A union bound over all parts, all small tuples, and all sign patterns shows that, for fixed d , taking $N = O(\log k)$ makes the total failure probability less than one. Hence a full target exists with

$$|V(H)| = kN = O(k \log k).$$

This is a standard application of Diestel's probabilistic method, Section 11.2: prove that the desired property holds with positive probability in an explicitly nonempty probability space. The random model proves existence; it is not a Monte Carlo approximation.

6 Lower-bound methods

6.1 Local lower bounds from 2-dipaths

Since $\chi_2(D) \leq \chi_o(D)$, any construction in which many pairs of vertices are adjacent or are endpoints of directed paths of length two forces many distinct colours.

Definition 6.1: Oriented clique

An oriented graph D is an oriented clique if

$$\chi_o(D) = |V(D)|.$$

Equivalently, every pair of distinct vertices is adjacent or is joined by a directed path of length two in one direction.

There exist oriented cliques on n vertices with only $O(n \log n)$ edges. This is striking: they are far sparser than complete graphs but still require a different oriented colour for every vertex.

6.2 Global counting lower bounds

Let an undirected graph G have n vertices and m edges. It has 2^m orientations. Suppose every orientation admitted an oriented k -colouring. Such an orientation could be encoded by

- a map of the n vertices into at most k colour classes;
- a tournament specifying the direction between the colour classes.

For a fixed map and fixed tournament, at most one orientation of every edge is compatible. If the total number of such encodings is less than 2^m , the pigeonhole principle produces an

orientation that is not k -colourable.

Wood's formal theorem implies that if an n -vertex graph has average degree at least $\log n$, then its oriented chromatic number is $\Omega(\sqrt{n})$. A stronger density assumption yields a lower bound in terms of $\sqrt{m}$. The exact algebra is less important than the message: once density reaches logarithmic order, a purely global counting argument already forces a large target.

6.3 Universal-target obstructions

For a class closed under disjoint union, one can choose several graphs that impose distinct structural requirements on any target. Their disjoint union forces a single universal target to satisfy all requirements simultaneously. The talk describes this as a third lower-bound method. The reported planar lower bound 19 was presented in this spirit; since it was described in the seminar as an unpublished improvement, these notes do not treat it as a fully verified published theorem.

7 Fixed degeneracy

7.1 The square-root scale

The one-subdivision of a clique gives

$$\chi_o(G) = \Omega(\sqrt{n})$$

already for 2-degenerate graphs. The speaker announced an unpublished upper bound of the form

$$\chi_o(G) = O(n^{1/2+o(1)})$$

for every fixed degeneracy d . The term $n^{o(1)}$ is subpolynomial; in the proof it is expected to arise from polylogarithmic losses. Thus the known construction and the announced upper bound match up to subpolynomial factors.

7.2 Proof architecture described in the seminar

The proof outline is as follows.

1. **Minimal counterexample.** Assume that G is a smallest graph not mapping to the chosen target. Minimality yields strong local structural restrictions.
2. **Low-/high-degree decomposition.** Degeneracy supplies low-degree vertices, but they are not coloured immediately. The proof separates a high-degree core from vertices intended for later extension.
3. **Extension graph.** Directed distance-two relations created through low-degree vertices are encoded by auxiliary degree-two vertices. The resulting graph resembles the subdivided-clique lower-bound examples.
4. **Transfer of colourability.** A suitable colouring of the extension graph controls all distance-two conflicts among high-degree vertices. One can then return to G and safely

extend the map to the low-degree vertices by using the full target property.

5. **Density contradiction.** If every local extension fails, each low-degree configuration forces many incidences in the extension graph. These local failures accumulate until the auxiliary graph has too many edges to remain compatible with the original degeneracy bound.

The speaker compared the final stage to discharging: no single local obstruction is immediately impossible, but the total collection of local obstructions exceeds the global sparsity budget.

8 Graphs on surfaces

8.1 Embeddings, faces, Euler characteristic, and Euler genus

Definition 8.1: Graph embedding in a surface

An embedding maps vertices to distinct points and edges to arcs whose interiors meet neither vertices nor other edge-arcs. The connected components of the complement are faces. The embedding is cellular if every face is an open disc.

Definition 8.2: Euler characteristic and Euler genus

For a cellular embedding with n vertices, m edges, and f faces,

$$n - m + f = \chi(S),$$

where $\chi(S)$ depends only on the surface. The Euler genus is

$$\text{eg}(S) := 2 - \chi(S),$$

so

$$n - m + f = 2 - \text{eg}(S).$$

Diestel's Appendix B, printed pages 423–427, develops surfaces, embeddings, Euler characteristic, handles, crosscaps, and Euler genus. Adding a handle increases Euler genus by two; adding a crosscap increases it by one. The Euler genus of a graph is the minimum Euler genus of a surface in which it embeds.

8.2 A Diestel-style comparison with ordinary colouring

For a simple cellularly embedded graph of Euler genus g , if every facial boundary has length at least three, then

$$m \leq 3n - 6 + 3g.$$

Diestel's Lemma 5.2.3 says that every k -chromatic graph has a k -chromatic subgraph of minimum degree at least $k - 1$. Applying the edge bound to such a critical core gives

$$(k - 1)n \leq 2m \leq 6n - 12 + 6g.$$

Since $n \geq k$, this yields a Heawood-type estimate of order $O(\sqrt{g})$ for ordinary chromatic number.

The mechanism is:

large ordinary chromatic number $\implies$ dense critical core,

whereas a fixed surface limits density.

This mechanism does not control oriented colouring. A one-subdivision may have minimum degree two and still have very large oriented chromatic number because the complexity is carried by directed paths of length two, not by a high-minimum-degree core.

8.3 Discrete curvature and discharging

For a triangulation of Euler genus g , Euler's formula and $3f = 2m$ imply

$$\sum_{v \in V} (6 - d(v)) = 12 - 6g.$$

The summands can be regarded as discrete vertex curvatures. A discharging proof assigns initial charges to vertices and/or faces and redistributes charge by local rules while preserving total charge. If the assumed absence of reducible configurations forces every final charge to have an impossible sign, one obtains a contradiction.

This is a discrete analogue of the Gauss–Bonnet principle: topology fixes the total curvature, while local geometry controls its distribution. Clow's proof uses more elaborate charges tailored to oriented-homomorphism extension; the displayed formula is the simplest conceptual model.

9 The main theorem for Euler genus

Define

$$\chi_o(g) := \max\{\chi_o(G) : \text{eg}(G) \leq g\}.$$

The main result discussed in the seminar is

$$\Omega\left(\frac{g}{\log g}\right) \leq \chi_o(g) \leq O(g \log g).$$

Thus the extremal oriented chromatic number for graphs of Euler genus at most g is nearly linear, up to logarithmic factors.

9.1 Lower bound

The lower-bound argument combines two facts.

1. There are oriented cliques on r vertices with only $O(r \log r)$ edges.
2. A connected graph with n vertices and m edges embeds in a surface of Euler genus at most $m - n + 1$.

For the second statement, embed a spanning tree on the sphere. Each non-tree edge can be inserted after adding at most one crosscap, and there are exactly $m - n + 1$ non-tree edges. Hence the sparse oriented clique has Euler genus $O(r \log r)$ but oriented chromatic number exactly r . Solving for r gives

$$r = \Omega\left(\frac{g}{\log g}\right).$$

9.2 Upper bound

The upper bound has the following architecture.

1. Construct a full oriented multipartite target with $k = \Theta(g)$ parts and $N = O(\log g)$ vertices per part.
2. Assume a smallest graph of Euler genus at most g does not map to the target.
3. Use minimality and the extension property to derive restrictive local conditions: low-degree vertices and their neighbourhoods must have very special forms.
4. Apply Euler's formula and a discharging argument to show that those local restrictions cannot all coexist in a graph of Euler genus g .
5. Conclude that no minimal counterexample exists.

The target has order

$$\Theta(g) \cdot O(\log g) = O(g \log g),$$

which yields the claimed upper bound.

The proof pattern to remember

The constants and individual degree thresholds are secondary. The conceptual synthesis is

probabilistic universal target + minimal-counterexample extension + Euler/discharging obstruction

The probabilistic method supplies a sufficiently rich target; minimality converts failed extensions into local structural restrictions; Euler genus and discharging show that the restrictions cannot coexist globally.

10 What the seminar contributes to broader mathematical training

10.1 Homomorphisms replace labels by structure-preserving maps

A proper colouring is not merely an assignment of integers: it is a map $G \rightarrow K_k$. Oriented colouring is a map $D \rightarrow H$ preserving more structure. The same conceptual move appears throughout mathematics:

- group representations $G \rightarrow GL(V)$;

- continuous maps between topological spaces;
- morphisms in algebraic geometry;
- homomorphisms to finite templates in constraint satisfaction.

The target records the compressed structure one wishes to retain.

10.2 Local solvability does not imply global compatibility

The failure of naive greedy colouring is a local-to-global phenomenon. Every individual step may look legal, yet a later vertex can reveal an incompatibility between two earlier choices. This is structurally analogous to transition-function compatibility on manifolds, gluing local PDE solutions, and cohomological obstructions: local data must satisfy global consistency conditions.

10.3 Different notions of sparsity answer different questions

Maximum degree, degeneracy, maximum average degree, and Euler genus are all notions of sparsity, but they control oriented colouring in different ways.

- Fixed maximum degree gives a finite bound depending on Δ , but the extremal dependence on Δ is exponential.
- Fixed degeneracy does not give a constant bound; the extremal value already grows as $\sqrt{n}$.
- Logarithmic average degree is sufficient for a counting argument to force $\Omega(\sqrt{n})$ colours.
- Euler genus combines a density restriction with topological structure, leading to a nearly linear function of g .

10.4 Fractional relaxation links combinatorics and convex analysis

Fractional colouring replaces an integral independent-set cover by a linear programme. The same route—discrete feasible set, convex relaxation, dual lower bound—underlies fractional matchings, fractional coverings, the Lovász theta function, and semidefinite relaxations. It is a direct bridge from graph theory to optimisation and convex analysis.

10.5 The probabilistic method is not Monte Carlo

Here probability proves the existence of a deterministic object:

$$\mathbb{P}(\text{the random target has all extension properties}) > 0.$$

Monte Carlo methods instead use random samples to approximate a numerical quantity. The same probability theory serves a different logical purpose.

10.6 Discharging is a discrete Gauss–Bonnet principle

Euler characteristic fixes the total charge; local degrees and face lengths determine its distribution. This viewpoint is useful beyond colouring, especially in topological graph theory, discrete differential geometry, geometric group theory, and discrete models of geometric PDE.

11 A focused reading path in Diestel

Stage 1: Essential vocabulary

1. **Diestel Section 1.1, printed pp. 2–3:** graphs, adjacency, independent sets, graph homomorphisms.
2. **Diestel Section 1.10, printed pp. 27–28:** directed graphs, orientations, oriented graphs.

Stage 2: Ordinary colouring as the baseline

1. **Diestel Sections 5.1–5.2, printed pp. 123–128:** the five-colour argument, greedy colouring, colouring number, and critical subgraphs.
2. **Chapter 5, Exercise 5(i), printed p. 148:** prove $\chi(G) \leq k$ if and only if $G \rightarrow K_k$.
3. **Chapter 5, Exercise 7:** examine a bad ordering for the greedy algorithm; this clarifies why the ordering matters even before the oriented setting is introduced.

Stage 3: Topological graph theory

1. **Diestel Section 4.2:** plane graphs and Euler’s formula.
2. **Appendix B, printed pp. 423–427:** surfaces, embeddings, Euler characteristic, Euler genus, handles, and crosscaps.

Stage 4: Probabilistic existence

1. **Diestel Sections 11.1–11.2, printed pp. 339–347:** random graphs and the probabilistic method.
2. Focus on the logical scheme “positive probability implies existence” and on the union bound for enforcing many local conditions simultaneously.

Stage 5: Research papers

1. Clow, *On Oriented Colourings of Graphs on Surfaces*: read the definitions first, then the lower-bound section, and finally the structure of the upper-bound argument.

2. Clow–Stacho, *Oriented Colouring Graphs of Bounded Degree and Degeneracy*: study the relation among degeneracy, 2-dipath colouring, and full targets.
3. Wood, *On the Oriented Chromatic Number of Dense Graphs*: study the counting lower-bound method.

12 Compact glossary

graph homomorphism

A vertex map preserving edges.

oriented graph

An orientation of a simple graph; no loops and no opposite pair of arcs.

tournament

An orientation of a complete graph.

oriented homomorphism

A vertex map preserving every arc and its direction.

oriented chromatic number

The minimum order of an oriented homomorphic target.

dichromatic number

The minimum number of acyclic induced parts in a vertex partition.

2-dipath colouring

A colouring separating adjacent vertices and endpoints of directed paths of length two.

degeneracy

The maximum, over all nonempty subgraphs, of their minimum degree.

maximum average degree

The maximum average degree over all nonempty subgraphs.

universal target

One target receiving a homomorphism from every graph in a class.

full target

A target realising every prescribed small in/out pattern.

oriented clique

An oriented graph whose oriented chromatic number equals its order.

Euler genus

The quantity $2 - \chi(S)$ for a surface.

discharging method

A proof method redistributing local charges while preserving their total.

13 Final synthesis

Three points to retain

1. Colouring is a homomorphism problem.

$$G \rightarrow K_k \quad \text{versus} \quad D \rightarrow T.$$

The ordinary target records which colours are distinct; an oriented target also records the permitted direction between every pair of colours.

2. Greedy colouring fails because constraints propagate to directed distance two.

A later vertex can reveal that two earlier equal colours force opposite arcs between the same pair of target colours. 2-dipath colourings and full targets repair the bookkeeping.

3. The surface theorem combines three toolkits.

probabilistic construction + minimal counterexample + Euler genus and discharging.

The result is

$$\Omega\left(\frac{g}{\log g}\right) \leq \chi_o(g) \leq O(g \log g).$$

References

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